



AI COMMERCE SYSTEM

Shopify

**An AI-guided way to build
the right shopping bundle**

A 10-slide system presentation for the AI Creator
Challenge

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September 2026

The shopping problem

Customers describe outcomes. Most stores only accept product keywords.

A customer may know the goal but not the product list

Examples include a comfortable work-from-home setup, a room refresh, a trip kit, or an outfit that works with what they already own.

1

Search friction

Keyword search asks customers to translate a real need into individual products.

2

Bundle risk

Separate purchases can miss compatibility, budget limits, stock, or an important requirement.

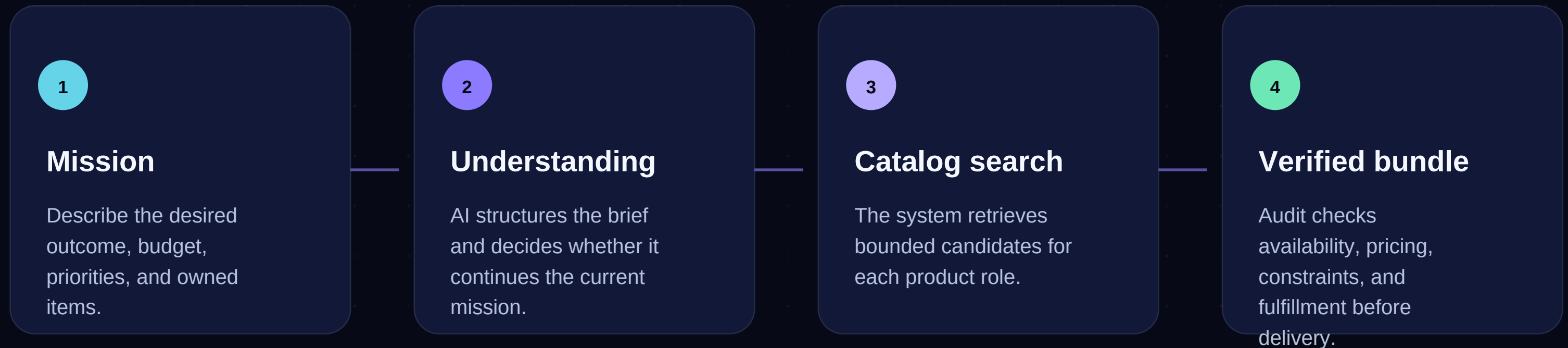
3

Trust gap

A recommendation loses value when the product facts and final price are not verified.

Shopy turns a goal into a verified shopping plan

The customer starts with a mission rather than a catalogue filter.



Result: a clear, customer-facing explanation plus product choices that passed deterministic checks.

Customer experience

One mission workspace supports text, voice, and image-based shopping.

EXAMPLE MISSION

**Build me a comfortable WFH setup
under RM2,000**

Budget RM 2,000

Priorities comfort, focus

Build for me

Text mission

A customer can state the desired result in natural language.

Voice input

Speech transcription fills the same mission flow so the customer can edit before running it.

Image shopping

Room, outfit, and object images can become a structured shopping brief and a bundle workspace.

System architecture

A full-stack

Next.js 16 + React 19

Storefront, mission workspace, chat, voice, image shopping

FastAPI + LangGraph

API routes, mission orchestration, planning, audit, brand voice

PostgreSQL + Redis

Commerce records, catalog facts, sessions, short-term shopping memory

Celery + email

Durable receipt delivery after paid orders

The browser has one origin. Docker Compose keeps PostgreSQL, Redis, the API, and the worker private to the application network.

AI application ability

Shopy uses AI where semantic judgement helps and deterministic code where product facts must remain authoritative.

QWEN

Primary intelligence layer

- Intent interpretation and mission planning
- Product-role query variants and bundle selection
- Vision analysis for rooms, outfits, and objects
- Customer-facing response writing

QWEN OMNI CAPTIONER

Voice to mission

Speech-to-text sends the customer recording into the same mission workflow. The application does not store the submitted audio.

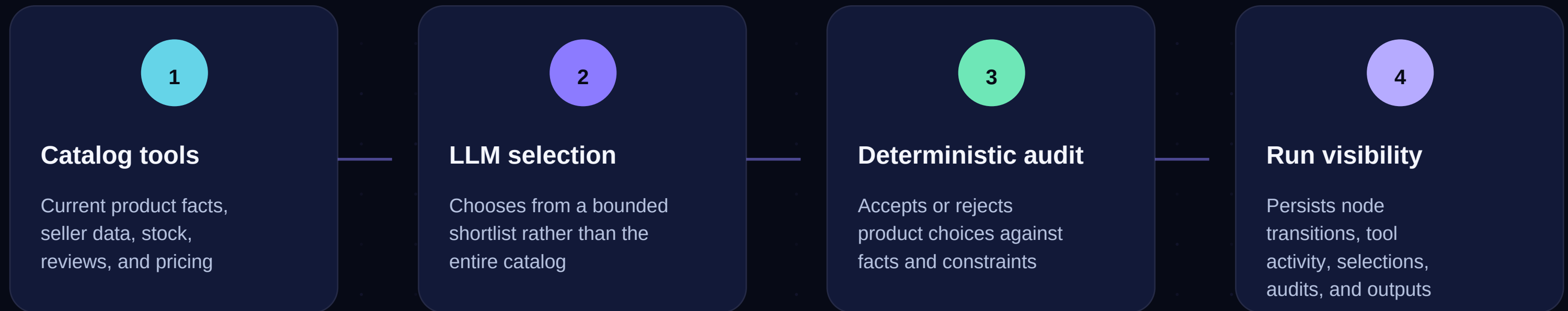
GEMINI

Configured fallback provider

Used only when Qwen fails and a Gemini API key is configured.

Trust, safety, and observability

The final response earns trust by separating AI judgement from catalog facts and audit rules.



No hidden chain-of-thought is stored. Raw vision image bytes are not persisted in orchestration logs.

Business value

Shopy improves the shopping journey by making the bundle, the rationale, and the final verification visible.



Higher mission completion

A goal-first input helps customers move beyond broad browsing and start with an intended outcome.



More confident bundles

Budget, ownership, constraints, availability, and fulfillment checks reduce avoidable decision friction.



Operational learning

Recorded run events show where the system searched, selected, audited, or failed so teams can improve it.

WHY IT IS WORTH DOING

The value comes from helping customers reach a purchase-ready plan while preserving factual controls that a normal chat response cannot guarantee.

Scale from here

The architecture already separates the experience, orchestration, catalog tools, memory, and observability needed for the next stage.

01 Catalog coverage

Connect more merchant catalogs while keeping retrieval bounded per product role.

02 Evaluation loop

Use audit outcomes, rejected choices, and run events to measure recommendation quality.

03 Personalization

Extend the session-scoped memory with consented preferences and repeat-purchase signals.

04 Operations

Run API and workers as independently scalable services while retaining queue-backed delivery.

Next milestone: prove quality and conversion with controlled live pilots before broad catalog expansion.

Evaluation summary

Shopy addresses the required questions and demonstrates a complete AI commerce system.

Innovation

A mission-first shopping workspace creates a structured bundle from a natural-language goal.

AI application ability

Qwen supports planning, vision, selection, and responses; Qwen Omni Captioner supports voice input.

Business value

A purchase-ready, verified plan can reduce search friction and make recommendations easier to trust.

Completeness

The system includes storefront, commerce API, catalog, memory, orchestration, audit, and delivery.

Presentation

A clear customer story connects the problem, the system, the safeguards, and the scale path.

Shopy

AI commerce built around the customer's goal

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